

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1131

Fork Pickett 100 kV
January 26, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1131. The Interconnection Customer proposes a 96.6 MW wind facility interconnecting at the Fork Pickett 100 kV substation in the Illinois area. The request was received on January 01, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 288 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,026,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Ft St Vrain-Alder Substation 500.0 kV line, the Exceksuir-Fork Pickett 100.0 kV loads to 127.0% of its 136 MVA rating. Project DFAX: 66.3%.

The Exceksuir-Gerald Eagle Electric 100.0 kV Line is overloaded to 120.2% of its 133 MVA emergency rating for the loss of the Ft St Vrain-Alder 500.0 kV. The project contributes 36.3% DFAX (35.1 MW) to this overload.

Ayer Rd-Gerald Eagle Electric 100.0 kV line: 109.8% loading (128 MVA) following the outage of the Fort St Vrain-Alder 500.0 kV; DFAX = 36.3%.

Waynedale-Montpelier Electric Generating Sta 230.0 kV: 106.4% loading (1,568 MVA) following the outage of the Richland Substation-Dekalb 500.0 kV; DFAX = 20.6%.

Swan Valley-Twin County Ranch 230.0 kV L: 104.9% loading (1,362 MVA) following the outage of the Troy Energy Llc Substation-Richland 500.0 kV L; DFAX = 17.0%.

The Wabash 500/230kV XFMR is overloaded to 104.0% of its 1,974 MVA emergency rating for the loss of the Richland-Dekalb 500.0 kV. The project contributes 19.1% DFAX (18.4 MW) to this overload.

The Waynedale-State Treet 230.0 kV Line is overloaded to 103.8% of its 1,242 MVA emergency rating for the loss of the Richland-Dekalb 500.0 kV. The project contributes 17.4% DFAX (16.8 MW) to this overload.

Richland-Twin County Ranch 230.0 kV: 103.5% loading (1,742 MVA) following the outage of the Troy Energy Llc-Richland 500.0 kV; DFAX = 19.0%.

Minnehaha-Napoleon Peaking Sta 230.0 kV: 103.3% loading (1,326 MVA) following the outage of the Troy Energy Llc-Richland 500.0 kV; DFAX = 12.8%.

Minnehaha-Station S Van Nuys 230.0 kV line: 101.2% loading (1,625 MVA) following the outage of the Troy Energy Llc-Richland 500.0 kV; DFAX = 16.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Exceksuir-Fork Pickett 100.0 kV	Rebuild line with higher-capacity conductor	\$1,816,000	78% (\$1,418,000)
Exceksuir-Gerald Eagle Electric 100.0 kV Line	Rebuild line with higher-capacity conductor	\$3,646,000	58% (\$2,133,000)
Gerald Eagle Electrc-Ayer Rd 100.0 kV Line	Reconductor line	\$2,610,000	58% (\$1,527,000)
Waynedale-Montpelier Electric Generating Sta 230.0 kV L	Reconductor line	\$21,611,000	1% (\$186,000)
Swan Valley-Twin County Ranch 230.0 kV L	Reconductor line	\$11,004,000	1% (\$95,000)
Wabash 500/230kV XFMR	Replace transformer with higher-rated unit	\$39,315,000	1% (\$338,000)
Waynedale-State Treet 230.0 kV Line	Reconductor line	\$5,276,000	1% (\$45,000)
Twin County Ranch-Richland 230.0 kV	Reconductor line	\$10,136,000	1% (\$87,000)
Napoleon Peaking Sta-Minnehaa 230.0 kV	Reconductor line	\$17,124,000	1% (\$147,000)
Sta S Van Nuys-Minnehaa 230.0 kV	Reconductor line	\$5,806,000	1% (\$50,000)
	Total allocated to project		\$6,026,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.